



Chapter 1

Introduction

In the evolving landscape of public administration, integrated digital systems have emerged as vital instruments for local government offices seeking to deliver more efficient, transparent, and responsive public services (David et al., 2023). In housing-related operations, such systems are especially valuable for organizing applicant records, monitoring beneficiaries, and streamlining workflows that are often slowed by fragmented, paper-based procedures. Szedmák et al. (2025) demonstrated that centralizing document management across municipalities significantly reduced processing times and standardized administrative workflows, while Andaya et al. (2025) similarly found that e-government platforms improved the responsiveness and efficiency of public services in the Philippine context.

Despite these gains, the pursuit of digital governance in the Philippines continues to face significant challenges. Andaya et al. (2025) noted that persistent barriers such as low digital literacy, inadequate ICT infrastructure, and institutional resistance constrain the potential of government information systems. Bithay et al. (2025) further emphasized that systems for local government units must be designed with interoperability and alignment with national digital governance standards in mind, while Pancho et al. (2025) stressed that effective implementation requires not only technological investment but also organizational support, continuous training, and coordination across government offices.



Previous research has demonstrated that digital platforms can meaningfully improve processes in public-sector settings. Uriawan et al. (2023) showed that a web-based housing management system improved coordination, information access, and transparency among stakeholders, while Erona et al. (2025) found that replacing fragmented manual records in a barangay local government unit with a digital system dramatically reduced data retrieval time and improved the overall quality of data management and decision-making. Dela Rosa (2022) similarly found that the absence of a structured digital system in a government agency created transparency issues and operational inefficiencies that were only resolved through the development of a web-based management information system.

In this context, the study focuses on developing an Integrated Housing Services and Monitoring System for the Talisay City Housing Authority. The system is intended to improve housing service delivery through digitization, centralized record management, and more efficient monitoring of applicants, beneficiaries, and housing units. This approach is expected to strengthen transparency, support data-driven planning, and enhance the delivery of housing-related services in the local government setting, especially at the Talisay Housing Authority.

1.1 Background of the Study

The Talisay City Housing Authority was established on December 8, 2018, during the administration of Mayor Neil Lizares, to formally address the housing needs of informal settler families in the city. Before its establishment, Talisay had no housing office, and relocation-related concerns were handled through the City Social Welfare and Development Office (CSWD) and executive assistants. Because of this



setup, housing-related records, applicant information, and relocation data were scattered and not centrally maintained in a single office.

When the Housing Authority was organized, its early operations involved gathering information, data, and applicant details from existing records and submissions, particularly those previously handled through the CSWD. This marked the beginning of the office's own housing record management and application handling. However, this initial consolidation remained manual, with records compiled via paper submissions, staff handling, and basic encoding rather than through a digital system.

In its early years, relocation activities were still limited. Only a few housing allocations were recorded in 2019, while more structured batch awarding became evident in 2020. Over time, the office assumed responsibility for profiling applicants, processing housing-related documents, preparing applicants for lot awarding, and monitoring relocation areas such as GK Cabaatangan. Despite this institutional development, the office continued to rely on physical records and manual workflows.

At present, the Talisay City Housing Authority handles several housing-related processes involving Informal Settler Families (ISF's). Applicants may enter the process through walk-in application. Once requirements are completed, applicants fill out forms, documents are routed for signatories, and records are physically filed. Priority applicants and walk-in applicants are managed separately, with walk-ins generally lined up on a first-in, first-out basis when lots are limited. Lot awarding is also dependent on actual lot availability.



After the assignment, monitoring of housing units remains manual. Staff rely on physical maps, block-and-lot records, caretaker reports, weekly site visits, and photo updates to determine whether units are occupied, vacant, abandoned, or under compliance concern. When beneficiaries fail to occupy or continue using their assigned units, notices are issued manually and followed up on, sometimes leading to repossession and blacklisting. Report generation is likewise labour-intensive because staff must manually compile data requested by management and partner offices, such as the Presidential Commission for the Urban Poor (PCUP), the Planning and Development Office (PDO), the CDRRMO, and the CSWD.

These practices show that, although the office has already centralized housing-related functions, its internal operations remain manual. As the volume of applicants, beneficiaries, unit records, and report requests increases, the current process becomes more difficult to sustain. Delays in retrieving records, difficulty tracking applicant and beneficiary status, and the absence of a centralized monitoring and reporting mechanism highlight the need for a system that not only digitizes files but also enhances and integrates the Authority's actual workflow.

1.2 Objectives of the Study

The main purpose of this study is to develop an Integrated Housing Services and Monitoring System for Local Government Human Settlements, specifically for the Talisay Housing Authority.

Specifically, this study aimed to:

1. To design and develop an Integrated Housing Services and Monitoring System with the following features:



- 1.1 Informal Settler Data Recording Management
 - 1.2 Housing Application and Evaluation
 - 1.3 Document Management Module
 - 1.4 Housing Unit and Occupancy Monitoring Module
 - 1.5 Case Management Module
 - 1.6 Analytics Dashboard
 - 1.7 SMS Notification System
 - 1.8 Caretaker Mobile Web Form
 - 1.9 Role-Based Access Control
2. To test the functionality of the developed system.
 3. To evaluate the software quality of the development system using ISO 25010:2011 and ISO 92411:2018 Criteria.
 4. To develop a User's Guide.

1.3 Significance of the Study

This study is significant as it proposes the development of an Integrated Housing Services and Monitoring System tailored for the Talisay City Housing Authority. The study will be beneficial to the following:

Talisay City Housing Authority. The system will provide the Authority with a centralized digital platform to manage informal settler family records, housing applications, housing unit availability, and case records. Through the Analytics and



Planning Dashboard, management can access real-time data summaries and generate reports that support informed decision-making and more effective planning of housing programs across the 27 barangays of Talisay City.

Administrative and Technical Staff of the Talisay City Housing Authority.

The system will reduce the Authority's reliance on manual, paper-based processes by digitizing household profiling, application tracking, document management, unit monitoring, and concern recording. Staff can retrieve information, update records, and monitor application and case statuses more efficiently, reducing administrative workload and minimizing errors caused by fragmented record-keeping.

Residents and Housing Applicants. The system will improve the transparency and organization of housing application processing and concern handling. By ensuring that applications and cases are properly recorded and monitored, the Authority can respond to resident inquiries more efficiently, thereby improving service delivery for informal settler families seeking housing assistance.

Management and Partner Offices. The system may support faster generation of summaries and reports requested by management and partner offices such as the PDO, CDRMO, CSWD, and PCUP by consolidating housing data into a searchable and report-ready format.

Future Researchers. The findings of this study may serve as a reference for researchers who intend to develop or improve systems related to housing management, local government service automation, and digital governance platforms for public sector institutions.

1.4 Scope and Limitation



This study focuses on the development of an Integrated Housing Services and Monitoring System for the Talisay City Housing Authority. The system is a web-based platform intended for the Authority's internal housing-related operations. The Informal Settler Recording Management feature records and manages applicant profiles. The Housing Application and Evaluation feature handles application processing from submission through lot awarding. The Document Management feature stores and retrieves applicant and beneficiary documents. The Housing Unit and Occupancy Monitoring feature tracks the occupancy status of housing units. The Case Management feature logs and monitors housing-related complaints and concerns. The Analytics Dashboard presents summarized housing data to support planning and reporting. Additionally, the system sends automated SMS notifications to applicants. A caretaker mobile form enables weekly occupancy reporting from the relocation site. For the technology stack, the proponents will use HTML5, CSS3, SCSS, and JavaScript for the front-end interface, with Python 3 and Django as the back-end framework. PostgreSQL serves as the database management system, with Git and GitHub used for version control throughout development.

While the system offers several helpful features, there were some limitations to consider. The study is limited to the internal workflow of the Talisay City Housing Authority and to authorized system users only. It does not include a public-facing portal or mobile application for applicants or residents. The system is intended to support office operations, but it does not replace external verification and decision processes performed by other offices or authorities. Specifically, it does not perform CDRRMO danger-zone certification, assessor verification of property ownership, sheriff-related processes for demolition cases, city engineering decisions on lot



measurements or boundary disputes, financial assistance processing by CSWD, or livelihood training management by PCUP. It also does not determine policy decisions such as the acquisition of new housing lots; rather, it only provides data that may support such decisions.

1.5 Definition of Terms

The following key terms have been operationally defined in relation to their specific usage in this study. These definitions aim to provide clarity and ensure a consistent understanding of concepts central to the study.

Analytics and Planning Dashboard. In this study, it refers to a visual interface within the system that presents summarized operational data, including ISF household counts per barangay, housing application status, housing unit availability, demographic breakdowns, and case summaries to support evidence-based planning decisions.

Beneficiary. In this study, it refers to an informal settler family whose housing application has been reviewed and approved by the Talisay City Housing Authority and is officially waiting for housing unit assignment.

Case Management Module. In this study, it refers to a system component that records, categorizes, and tracks housing-related concerns or complaints raised by residents or beneficiaries, allowing staff to monitor their status and resolution.

Document Management Module. In this study, it refers to a system component that records, archives, and organizes documents submitted by housing applicants and allows staff to search and filter records per applicant.



Housing Application and Beneficiary Management Module. In this study, it refers to a system component that records and monitors housing applications submitted by informal settler families, tracks their status through the evaluation process, and manages approved applicants as beneficiaries waiting for unit assignment.

Housing Unit and Occupancy Monitoring Module. In this study, it refers to a system component that tracks all housing units across the three areas of GK Cabatangan, monitors their current occupancy status, supports beneficiary unit assignment, and allows staff to flag units for compliance concerns after assignment.

Informal Settler Family. In this study, it refers to a household residing on land or in an area that it does not legally own, including those occupying private property, government property, easements, riverside areas, or danger zones subject to relocation.

ISF Management Module. In this study, it refers to the system component that records and maintains household profiles of informal settler families, including their demographic information, applicant source, and priority classification.

Integrated Housing Services and Monitoring System. In this study, it refers to the web-based system developed for the Talisay City Housing Authority that integrates informal settler family monitoring, housing application and beneficiary management, document management, housing unit and occupancy monitoring, case management, and analytics-driven planning support into a single platform.



Occupancy Status. In this study, it refers to the current condition of a housing unit as recorded in the system, which may be classified as available, occupied, vacant, abandoned, or flagged for a compliance concern.

Priority List. In this study, it refers to an applicant or household given urgent consideration based on verified conditions such as danger-zone residency, landowner endorsement, demolition-related circumstances, or other officially recognized bases used by the office.

Role-Based Access Control. In this study, it refers to a system security feature that restricts access and permissions according to the user's assigned role within the system.

Standby Queue. In this study, it refers to the recorded list of applicants or beneficiaries who are already processed or prioritized but are still waiting because no lot or housing unit is currently available.

Walk-in Applicant. In this study, it refers to an informal settler family that approaches the Talisay City Housing Authority on their own initiative to request housing assistance and is recorded in a separate waiting list from priority referrals